

Q.7: Best Fit Allocation

```
#include <iostream>
using namespace std;

int main() {
    int m, n;

    cout << "Enter number of memory blocks: ";
    cin >> m;

    int blockSize[20];
    cout << "Enter size of each block:\n";
    for (int i = 0; i < m; i++) {
        cout << " Block " << i + 1 << ": ";
        cin >> blockSize[i];
    }

    cout << "\nEnter number of processes: ";
    cin >> n;

    int processSize[20];
    cout << "Enter size of each process:\n";
    for (int i = 0; i < n; i++) {
        cout << " Process " << i + 1 << ": ";
        cin >> processSize[i];
    }

    int allocation[20];
    for (int i = 0; i < n; i++)
        allocation[i] = -1;

    for (int i = 0; i < n; i++) {
        int bestIndex = -1;

        for (int j = 0; j < m; j++) {
            if (blockSize[j] >= processSize[i]) {
                if (bestIndex == -1 || blockSize[j] < blockSize[bestIndex])
                    bestIndex = j;
            }
        }

        if (bestIndex != -1) {
            allocation[i] = bestIndex;
            blockSize[bestIndex] = blockSize[bestIndex] - processSize[i];
        }
    }

    cout << "\nProcess\tSize\tBlock Allotted\n";
    for (int i = 0; i < n; i++) {
        cout << "P" << i + 1 << "\t" << processSize[i] << "\t";
        if (allocation[i] != -1)
            cout << "Block " << allocation[i] + 1 << endl;
    }
}
```

```

        else
            cout << "Not Allotted" << endl;
        }

    return 0;
}

```

Output:

Problems

Output

Debug Console

Terminal

Ports

```

cd "/Users/pratikthapamagar/College/OS_lab_sem4/" && g++ lab_7.cpp -o lab_7 && "/Users/pratikthapamagar/College/OS_lab_sem4/"lab_7
pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 % cd "/Users/pratikthapamagar/College/OS_lab_sem4/" && g++ lab_7.cpp -o lab_7 && "/Users/pratikthapamagar/College/OS_lab_sem4/"lab_7
Enter number of memory blocks: 5
Enter size of each block:
    Block 1: 100
    Block 2: 500
    Block 3: 200
    Block 4: 300
    Block 5: 600

Enter number of processes: 2
Enter size of each process:
    Process 1: 212
    Process 2: 417

Process Size      Block Allotted
P1          212    Block 4
P2          417    Block 2
pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 %

```

Q.8: Worst Fit Allocation

```
#include <iostream>
using namespace std;

int main() {
    int m, n;

    cout << "Enter number of memory blocks: ";
    cin >> m;

    int blockSize[20];
    cout << "Enter size of each block:\n";
    for (int i = 0; i < m; i++) {
        cout << " Block " << i + 1 << ": ";
        cin >> blockSize[i];
    }

    cout << "\nEnter number of processes: ";
    cin >> n;

    int processSize[20];
    cout << "Enter size of each process:\n";
    for (int i = 0; i < n; i++) {
        cout << " Process " << i + 1 << ": ";
        cin >> processSize[i];
    }

    int allocation[20];
    for (int i = 0; i < n; i++)
        allocation[i] = -1;

    for (int i = 0; i < n; i++) {
        int worstIndex = -1;

        for (int j = 0; j < m; j++) {
            if (blockSize[j] >= processSize[i]) {
                if (worstIndex == -1 || blockSize[j] > blockSize[worstIndex])
                    worstIndex = j;
            }
        }

        if (worstIndex != -1) {
            allocation[i] = worstIndex;
            blockSize[worstIndex] = blockSize[worstIndex] - processSize[i];
        }
    }

    cout << "\nProcess\tSize\tBlock Allotted\n";
    for (int i = 0; i < n; i++) {
        cout << "P" << i + 1 << "\t" << processSize[i] << "\t";
        if (allocation[i] != -1)
```

```

        cout << "Block " << allocation[i] + 1 << endl;
    else
        cout << "Not Allotted" << endl;
}

return 0;
}

```

Output:

```

Pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 % cd "/Users/pratikthapamagar/College/OS_lab_sem4/" && g++ lab_8.cpp -o lab_8 && "/Users/pratikthapamagar/College/OS_lab_sem4/"lab_8
Enter number of memory blocks: 5
Enter size of each block:
  Block 1: 100
  Block 2: 500
  Block 3: 200
  Block 4: 300
  Block 5: 600

Enter number of processes: 2
Enter size of each process:
  Process 1: 212
  Process 2: 417

Process Size    Block Allotted
P1      212     Block 5
P2      417     Block 2
Pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 %

```

Q.9: First Fit Allocation

```
#include <iostream>
using namespace std;

int main() {
    int m, n;

    cout << "Enter number of memory blocks: ";
    cin >> m;

    int blockSize[20];
    cout << "Enter size of each block:\n";
    for (int i = 0; i < m; i++) {
        cout << " Block " << i + 1 << ": ";
        cin >> blockSize[i];
    }

    cout << "\nEnter number of processes: ";
    cin >> n;

    int processSize[20];
    cout << "Enter size of each process:\n";
    for (int i = 0; i < n; i++) {
        cout << " Process " << i + 1 << ": ";
        cin >> processSize[i];
    }

    int allocation[20];
    for (int i = 0; i < n; i++)
        allocation[i] = -1;

    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            if (blockSize[j] >= processSize[i]) {
                allocation[i] = j;
                blockSize[j] = blockSize[j] - processSize[i];
                break;
            }
        }
    }

    cout << "\nProcess\tSize\tBlock Allotted\n";
    for (int i = 0; i < n; i++) {
        cout << "P" << i + 1 << "\t" << processSize[i] << "\t";
        if (allocation[i] != -1)
            cout << "Block " << allocation[i] + 1 << endl;
        else
            cout << "Not Allotted" << endl;
    }
    return 0;
}
```

Output:

```
Problems  Output  Debug Console  Terminal  Ports

● pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 % cd "/Users/pratikthapamagar/College/OS_lab_sem4/" && g++ lab_9.cpp -o lab_9 && "/Users/pratikthapamagar/College/OS_lab_sem4/"lab_9
Enter number of memory blocks: 5
Enter size of each block:
  Block 1: 100
  Block 2: 500
  Block 3: 200
  Block 4: 300
  Block 5: 600

Enter number of processes: 2
Enter size of each process:
  Process 1: 212
  Process 2: 417

Process Size    Block Allotted
P1      212     Block 2
P2      417     Block 5
○ pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 %
```

Q.10: FCFS Disk Scheduling

```
#include <iostream>
using namespace std;

int main()
{
    int n, totalCylinders, head;

    cout << "Enter total number of cylinders (disk size): ";
    cin >> totalCylinders;

    cout << "Enter current head position: ";
    cin >> head;

    cout << "Enter number of pending requests: ";
    cin >> n;

    int request[50];
    cout << "Enter the request queue:\n";
    for (int i = 0; i < n; i++) {
        cout << " Request " << i + 1 << ": ";
        cin >> request[i];
    }

    int totalSeekTime = 0;
    int currentPos = head;

    cout << "\nSeek Sequence: " << currentPos;

    for (int i = 0; i < n; i++) {
        totalSeekTime = totalSeekTime + abs(currentPos - request[i]);
        currentPos = request[i];
        cout << " -> " << currentPos;
    }

    cout << "\n\nTotal Seek Time = " << totalSeekTime << endl;
    cout << "Final position of Read/Write head = " << currentPos << endl;

    return 0;
}
```

Output:

```
Problems  Output  Debug Console  Terminal  Ports

● pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 % cd "/Users/pratikthapamagar/College
/OS_lab_sem4/" && g++ lab_10.cpp -o lab_10 && "/Users/pratikthapamagar/College/OS_lab_
sem4/"lab_10
Enter total number of cylinders (disk size): 150
Enter current head position: 30
Enter number of pending requests: 10
Enter the request queue:
  Request 1: 35
  Request 2: 70
  Request 3: 45
  Request 4: 15
  Request 5: 65
  Request 6: 20
  Request 7: 80
  Request 8: 90
  Request 9: 75
  Request 10: 130

Seek Sequence: 30 -> 35 -> 70 -> 45 -> 15 -> 65 -> 20 -> 80 -> 90 -> 75 -> 130

Total Seek Time = 330
Final position of Read/Write head = 130
○ pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 %
```


Q.11: SCAN Disk Scheduling

```
#include <iostream>
using namespace std;

int main()
{
    int n, totalCylinders, head, direction;

    cout << "Enter total number of cylinders (disk size): ";
    cin >> totalCylinders;

    cout << "Enter current head position: ";
    cin >> head;

    cout << "Enter number of pending requests: ";
    cin >> n;

    int request[50];
    cout << "Enter the request queue:\n";
    for (int i = 0; i < n; i++) {
        cout << " Request " << i + 1 << ": ";
        cin >> request[i];
    }

    cout << "Enter head movement direction (0 = towards lower cylinders, 1 = towards higher cylinders): ";
    cin >> direction;

    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (request[j] > request[j + 1]) {
                int temp = request[j];
                request[j] = request[j + 1];
                request[j + 1] = temp;
            }
        }
    }

    int left[50], right[50];
    int leftCount = 0, rightCount = 0;

    for (int i = 0; i < n; i++) {
        if (request[i] < head)
            left[leftCount++] = request[i];
        else
            right[rightCount++] = request[i];
    }

    int totalSeekTime = 0;
    int currentPos = head;

    cout << "\nSeek Sequence: " << currentPos;
```

```

if (direction == 1) {
    for (int i = 0; i < rightCount; i++) {
        totalSeekTime = totalSeekTime + abs(currentPos - right[i]);
        currentPos = right[i];
        cout << " -> " << currentPos;
    }

    if (currentPos != totalCylinders - 1) {
        totalSeekTime = totalSeekTime + abs(currentPos - (totalCylinders - 1));
        currentPos = totalCylinders - 1;
        cout << " -> " << currentPos;
    }

    for (int i = leftCount - 1; i >= 0; i--) {
        totalSeekTime = totalSeekTime + abs(currentPos - left[i]);
        currentPos = left[i];
        cout << " -> " << currentPos;
    }
} else {
    for (int i = leftCount - 1; i >= 0; i--) {
        totalSeekTime = totalSeekTime + abs(currentPos - left[i]);
        currentPos = left[i];
        cout << " -> " << currentPos;
    }

    if (currentPos != 0) {
        totalSeekTime = totalSeekTime + abs(currentPos - 0);
        currentPos = 0;
        cout << " -> " << currentPos;
    }

    for (int i = 0; i < rightCount; i++) {
        totalSeekTime = totalSeekTime + abs(currentPos - right[i]);
        currentPos = right[i];
        cout << " -> " << currentPos;
    }
}

cout << "\n\nTotal Seek Time = " << totalSeekTime << endl;
cout << "Final position of Read/Write head = " << currentPos << endl;

return 0;
}

```

Output:

```
Problems  Output  Debug Console  Terminal  Ports

● pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 % cd "/Users/pratikthapamagar/College
/OS_lab_sem4/" && g++ lab_11.cpp -o lab_11 && "/Users/pratikthapamagar/College/OS_lab_
sem4/"lab_11
Enter total number of cylinders (disk size): 200
Enter current head position: 80
Enter number of pending requests: 10
Enter the request queue:
  Request 1: 23
  Request 2: 45
  Request 3: 67
  Request 4: 156
  Request 5: 12
  Request 6: 190
  Request 7: 134
  Request 8: 54
  Request 9: 178
  Request 10: 98
Enter head movement direction (0 = towards lower cylinders, 1 = towards higher cylinde
rs): 0

Seek Sequence: 80 -> 67 -> 54 -> 45 -> 23 -> 12 -> 0 -> 98 -> 134 -> 156 -> 178 -> 190

Total Seek Time = 270
Final position of Read/Write head = 190
○ pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 %
```

Q.12: LOOK Disk Scheduling

```
#include <iostream>
using namespace std;

int main()
{
    int n, totalCylinders, head, direction;

    cout << "Enter total number of cylinders (disk size): ";
    cin >> totalCylinders;

    cout << "Enter current head position: ";
    cin >> head;

    cout << "Enter number of pending requests: ";
    cin >> n;

    int request[50];
    cout << "Enter the request queue:\n";
    for (int i = 0; i < n; i++) {
        cout << " Request " << i + 1 << ": ";
        cin >> request[i];
    }

    cout << "Enter head movement direction (0 = towards lower cylinders, 1 = towards higher cylinders): ";
    cin >> direction;

    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (request[j] > request[j + 1]) {
                int temp = request[j];
                request[j] = request[j + 1];
                request[j + 1] = temp;
            }
        }
    }

    int left[50], right[50];
    int leftCount = 0, rightCount = 0;

    for (int i = 0; i < n; i++) {
        if (request[i] < head)
            left[leftCount++] = request[i];
        else
            right[rightCount++] = request[i];
    }

    int totalSeekTime = 0;
    int currentPos = head;

    cout << "\nSeek Sequence: " << currentPos;
```

```

if (direction == 1) {
    for (int i = 0; i < rightCount; i++) {
        totalSeekTime = totalSeekTime + abs(currentPos - right[i]);
        currentPos = right[i];
        cout << " -> " << currentPos;
    }

    for (int i = leftCount - 1; i >= 0; i--) {
        totalSeekTime = totalSeekTime + abs(currentPos - left[i]);
        currentPos = left[i];
        cout << " -> " << currentPos;
    }
} else {
    for (int i = leftCount - 1; i >= 0; i--) {
        totalSeekTime = totalSeekTime + abs(currentPos - left[i]);
        currentPos = left[i];
        cout << " -> " << currentPos;
    }

    for (int i = 0; i < rightCount; i++) {
        totalSeekTime = totalSeekTime + abs(currentPos - right[i]);
        currentPos = right[i];
        cout << " -> " << currentPos;
    }
}

cout << "\n\nTotal Seek Time = " << totalSeekTime << endl;
cout << "Final position of Read/Write head = " << currentPos << endl;

return 0;
}

```

Output:

```
Problems  Output  Debug Console  Terminal  Ports

● pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 % cd "/Users/pratikthapamagar/College
/OS_lab_sem4/" && g++ lab_12.cpp -o lab_12 && "/Users/pratikthapamagar/College/OS_lab_
sem4/"lab_12
Enter total number of cylinders (disk size): 200
Enter current head position: 80
Enter number of pending requests: 10
Enter the request queue:
  Request 1: 23
  Request 2: 45
  Request 3: 67
  Request 4: 156
  Request 5: 12
  Request 6: 190
  Request 7: 134
  Request 8: 54
  Request 9: 178
  Request 10: 98
Enter head movement direction (0 = towards lower cylinders, 1 = towards higher cylinde
rs): 0

Seek Sequence: 80 -> 67 -> 54 -> 45 -> 23 -> 12 -> 98 -> 134 -> 156 -> 178 -> 190

Total Seek Time = 246
Final position of Read/Write head = 190
○ pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 %
```

Q.13: Deadlock Detection (Safety Algorithm)

```
#include <iostream>
using namespace std;

int main() {
    int p, r;

    cout << "Enter number of processes: ";
    cin >> p;

    cout << "Enter number of resource types: ";
    cin >> r;

    int allocation[10][10];
    int maxDemand[10][10];
    int need[10][10];
    int available[10];

    cout << "\nEnter Allocation matrix:\n";
    for (int i = 0; i < p; i++) {
        cout << " Process " << i << ":\n";
        for (int j = 0; j < r; j++) {
            cout << " Allocation of resource " << j << " : ";
            cin >> allocation[i][j];
        }
    }

    cout << "\nEnter Max matrix:\n";
    for (int i = 0; i < p; i++) {
        cout << " Process " << i << ":\n";
        for (int j = 0; j < r; j++) {
            cout << " Max demand of resource " << j << " : ";
            cin >> maxDemand[i][j];
        }
    }

    cout << "\nEnter Available resources:\n";
    for (int j = 0; j < r; j++) {
        cout << " Resource " << j << " : ";
        cin >> available[j];
    }

    for (int i = 0; i < p; i++) {
        for (int j = 0; j < r; j++) {
            need[i][j] = maxDemand[i][j] - allocation[i][j];
        }
    }

    bool finished[10];
    for (int i = 0; i < p; i++)
        finished[i] = false;
```

```

int safeSequence[10];
int count = 0;

int work[10];
for (int j = 0; j < r; j++)
    work[j] = available[j];

while (count < p) {
    bool foundProcess = false;

    for (int i = 0; i < p; i++) {
        if (finished[i])
            continue;

        bool canRun = true;
        for (int j = 0; j < r; j++) {
            if (need[i][j] > work[j]) {
                canRun = false;
                break;
            }
        }

        if (canRun) {
            for (int j = 0; j < r; j++)
                work[j] = work[j] + allocation[i][j];

            finished[i] = true;
            safeSequence[count] = i;
            count++;
            foundProcess = true;
        }
    }

    if (!foundProcess)
        break;
}

if (count == p) {
    cout << "\nThe system is in a SAFE state (no deadlock).\n";
    cout << "Safe Sequence: ";
    for (int i = 0; i < count; i++) {
        cout << "P" << safeSequence[i];
        if (i != count - 1)
            cout << "-> ";
    }
    cout << endl;
} else {
    cout << "\nThe system is in an UNSAFE state (deadlock may occur).\n";
}

return 0;
}

```


Output:

```
Problems  Output  Debug Console  Terminal  Ports

● pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 % cd "/Users/pratikthapamagar/College/OS_lab_sem4/" && g++ lab_13.cpp -o lab_13 && "/Users/pratikthapamagar/College/OS_lab_sem4/"lab_13
Enter number of processes: 5
Enter number of resource types: 3

Enter Allocation matrix:
Process 0:
  Allocation of resource 0: 0
  Allocation of resource 1: 1
  Allocation of resource 2: 0
Process 1:
  Allocation of resource 0: 2
  Allocation of resource 1: 0
  Allocation of resource 2: 0
Process 2:
  Allocation of resource 0: 3
  Allocation of resource 1: 0
  Allocation of resource 2: 2
Process 3:
  Allocation of resource 0: 2
  Allocation of resource 1: 1
  Allocation of resource 2: 1
Process 4:
  Allocation of resource 0: 0
  Allocation of resource 1: 0
  Allocation of resource 2: 2

Enter Max matrix:
Process 0:
  Max demand of resource 0: 7
  Max demand of resource 1: 5
  Max demand of resource 2: 3
Process 1:
  Max demand of resource 0: 3
  Max demand of resource 1: 2
  Max demand of resource 2: 2
Process 2:
  Max demand of resource 0: 9
  Max demand of resource 1: 0
  Max demand of resource 2: 2
Process 3:
  Max demand of resource 0: 2
  Max demand of resource 1: 2
  Max demand of resource 2: 2
Process 4:
  Max demand of resource 0: 4
  Max demand of resource 1: 3
  Max demand of resource 2: 3

Enter Available resources:
Resource 0: 3
Resource 1: 3
Resource 2: 2

The system is in a SAFE state (no deadlock).
Safe Sequence: P1 -> P3 -> P4 -> P0 -> P2
○ pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 %
```